

海洋資料分析實作

以海洋熱浪（Marine Heatwaves）為例 🌊🔥

葉庭禎



美國南加州

US news

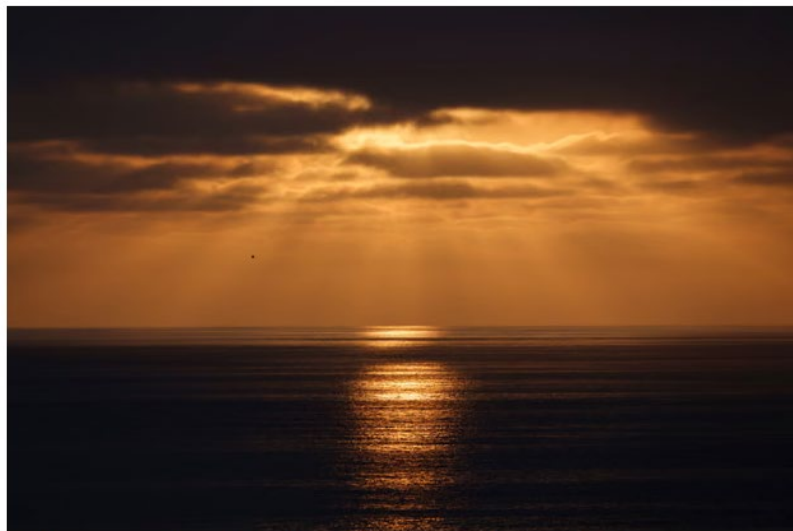
Record high ocean temperatures off southern California raise fears of prolonged marine heatwave

Researchers warn the high-pressure conditions could disrupt marine life and ecosystems if it continues

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Roque Planas

Sat 4 Apr 2026 13.00 BST

[Share](#)[Prefer the Guardian on Google](#)

📷 The waters of southern California historically warm every few years. But the marine heatwave that started last fall wasn't caused by tropical currents. Photograph: Kevin Carter/Getty Images

這篇新聞，有哪些地方值得我們注意？

- Record high ocean temperatures
 - record 是跟誰比？
 - 多久的紀錄？
 - 哪個區域？
- 10°F above historical average
 - Threshold 會隨一年四季改變
 - 去年平均？
 - 30 年平均？
 - 哪一段時間的平均？
- Marine heatwave
 - 定義？海溫高就算嗎？

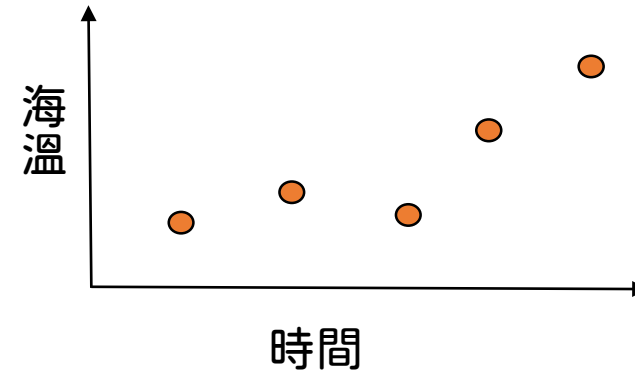
The Guardian (2026)

<https://www.theguardian.com/us-news/2026/apr/04/high-ocean-temperatures-california-marine-heatwave>

- Record high ocean temperatures ³
- historical average
- MHW

• 思考資料呈現方式

地圖



• 哪一種資料？

- 每小時/每日？
- 每月平均？
- 每年平均？

• 哪一個地點？

- La Jolla
- Southern California

- 需要：
 - Daily SST (sea surface temperature)
 - Southern California
 - 時間序列
- GHR SST data (Group for High Resolution Sea Surface Temperature)
 - NODASS 資料包 (2025-12 一個月)
 - Hidy Viewer 圖台 (2023-01-01 至今)

Hidy Viewer 

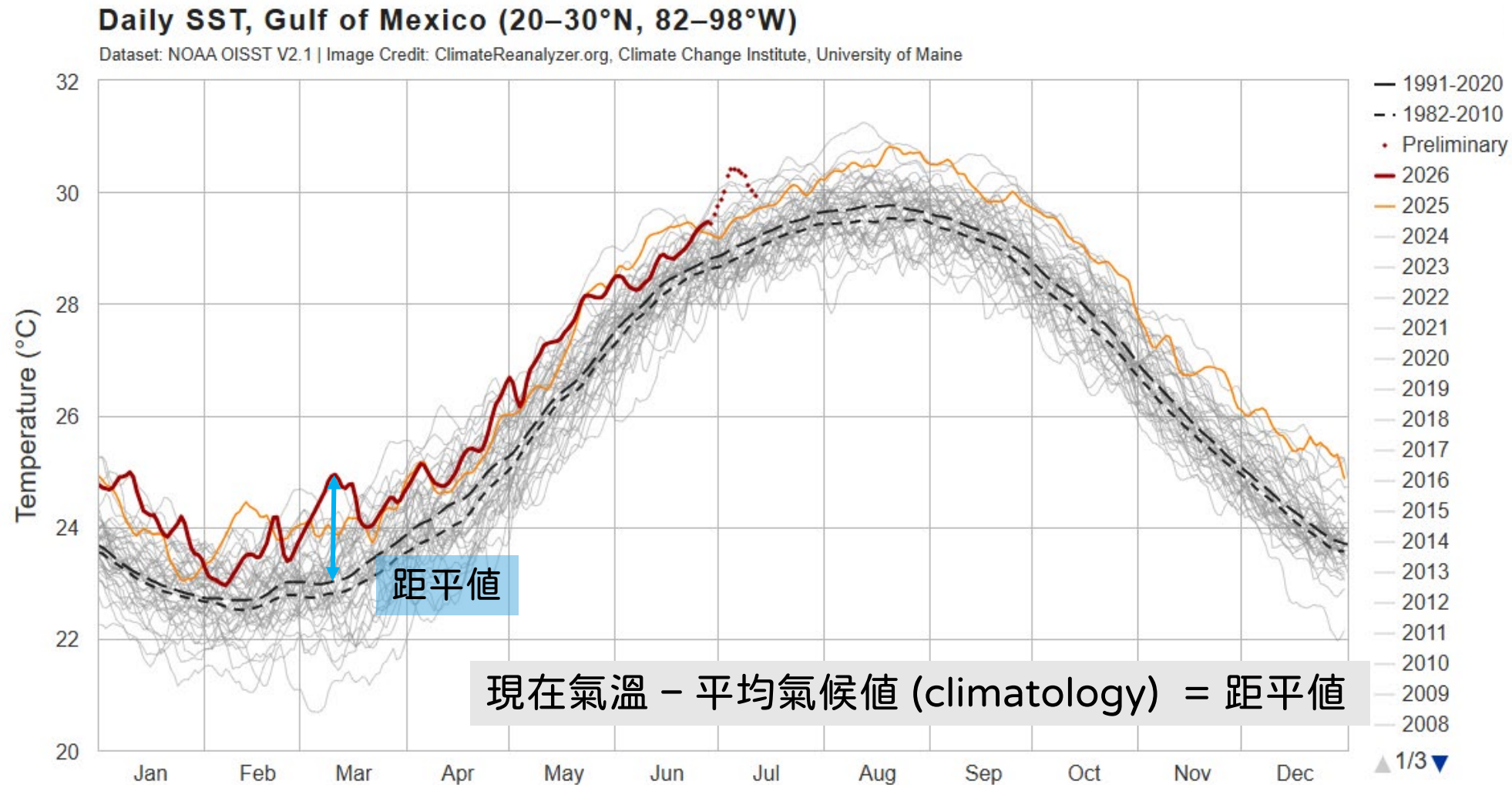
[**https://odbview.oc.ntu.edu.tw/hidy/**](https://odbview.oc.ntu.edu.tw/hidy/)

- 請利用 Hidy Viewer
 - 衛星圖像 > 最新圖資 > 海表溫度
 - 驗證新聞中的這句話：
 - *Over the last three months, several stations have repeatedly posted record-breaking daily high temperatures*
 - (2026-04-04)

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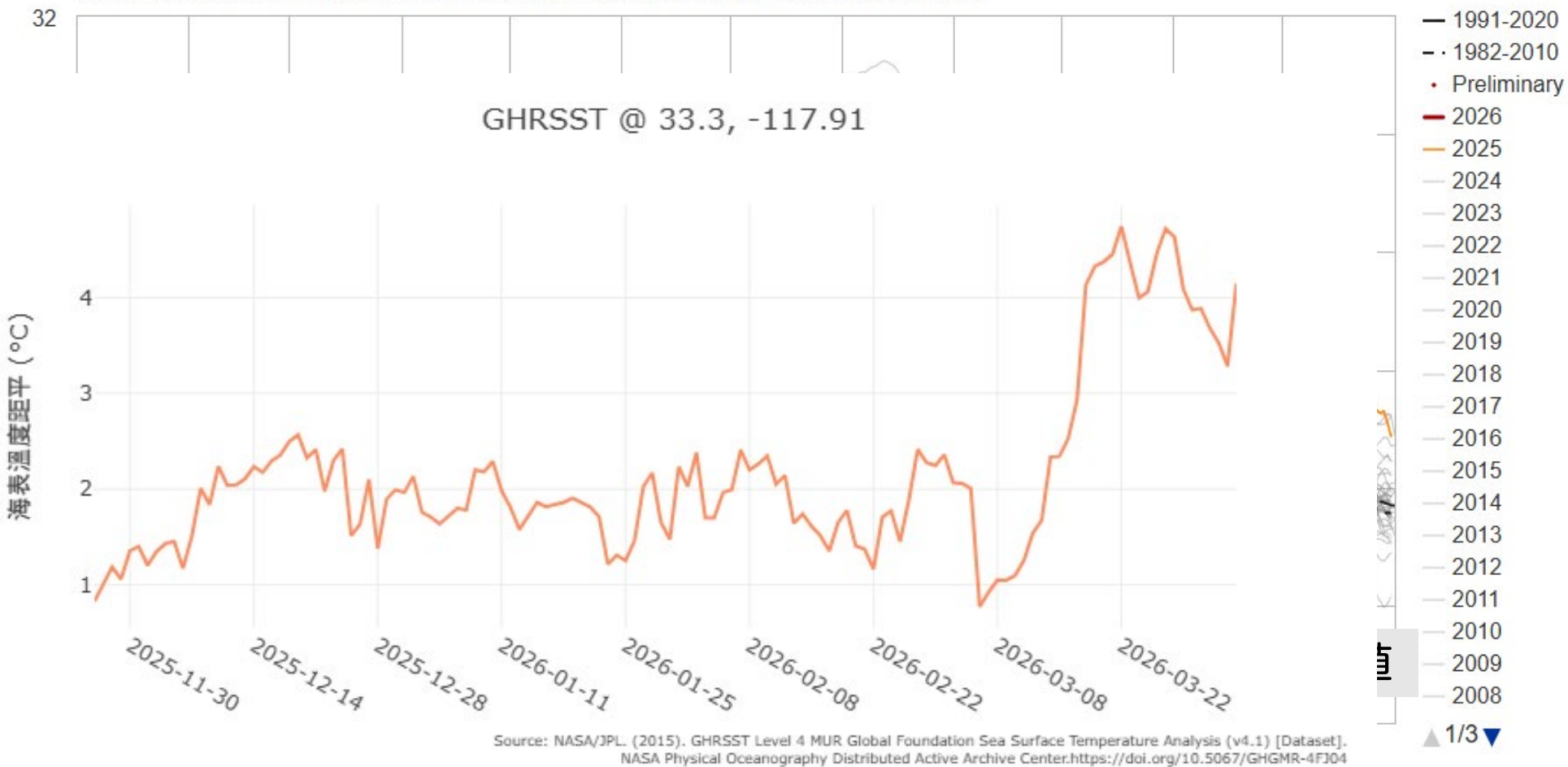
比平常熱多少？

平均氣候值
(climatology)

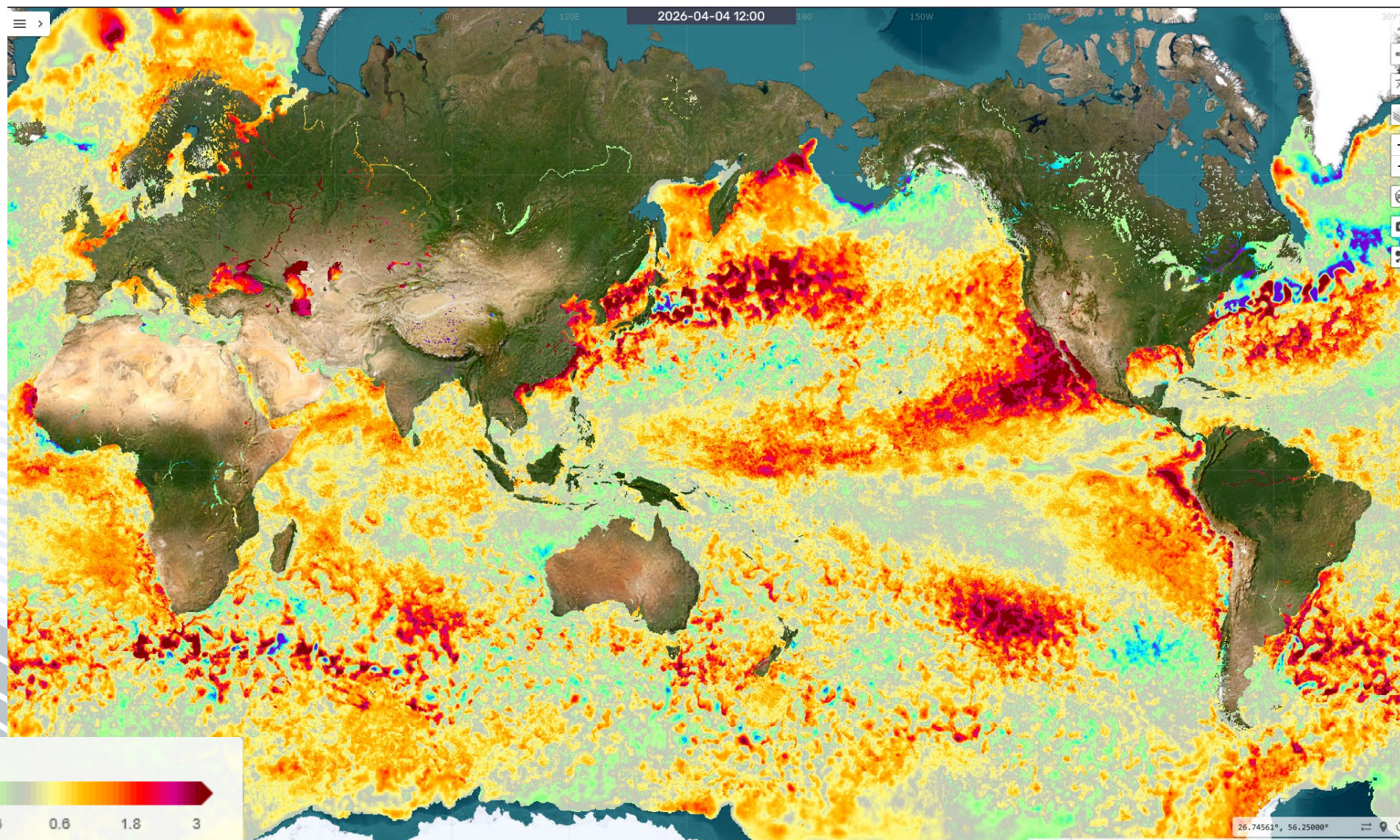
比平常熱多少？

Daily SST, Gulf of Mexico (20–30°N, 82–98°W)

Dataset: NOAA OISST V2.1 | Image Credit: ClimateReanalyzer.org, Climate Change Institute, University of Maine

平均氣候值
(climatology)

Anomaly > 0 ?? Marine Heatwave



[https://odbview.oc.ntu.edu.tw/hidy/?webmap=identifier:ssta&map=z:3;c:\[10.4878,156.2695\];datetime:2026-04-04T04:00:00.000Z](https://odbview.oc.ntu.edu.tw/hidy/?webmap=identifier:ssta&map=z:3;c:[10.4878,156.2695];datetime:2026-04-04T04:00:00.000Z)

Marine Heatwaves 🌊 🔥



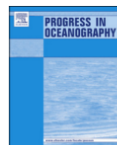
Progress in Oceanography 141 (2016) 227–238



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A hierarchical approach to defining marine heatwaves

Alistair J. Hobday^{a,*}, Lisa V. Alexander^{b,c}, Sarah E. Perkins^{b,c}, Dan A. Smale^{d,e}, Sandra C. Straub^e, Eric C.J. Oliver^{b,f}, Jessica A. Benthuyssen^g, Michael T. Burrows^h, Markus G. Donat^{b,c}, Ming Fengⁱ, Neil J. Holbrook^{b,f}, Pippa J. Moore^j, Hillary A. Scannell^{k,l}, Alex Sen Gupta^{b,c}, Thomas Wernberg^e



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ARTICLE INFO

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Received 11 June 2015

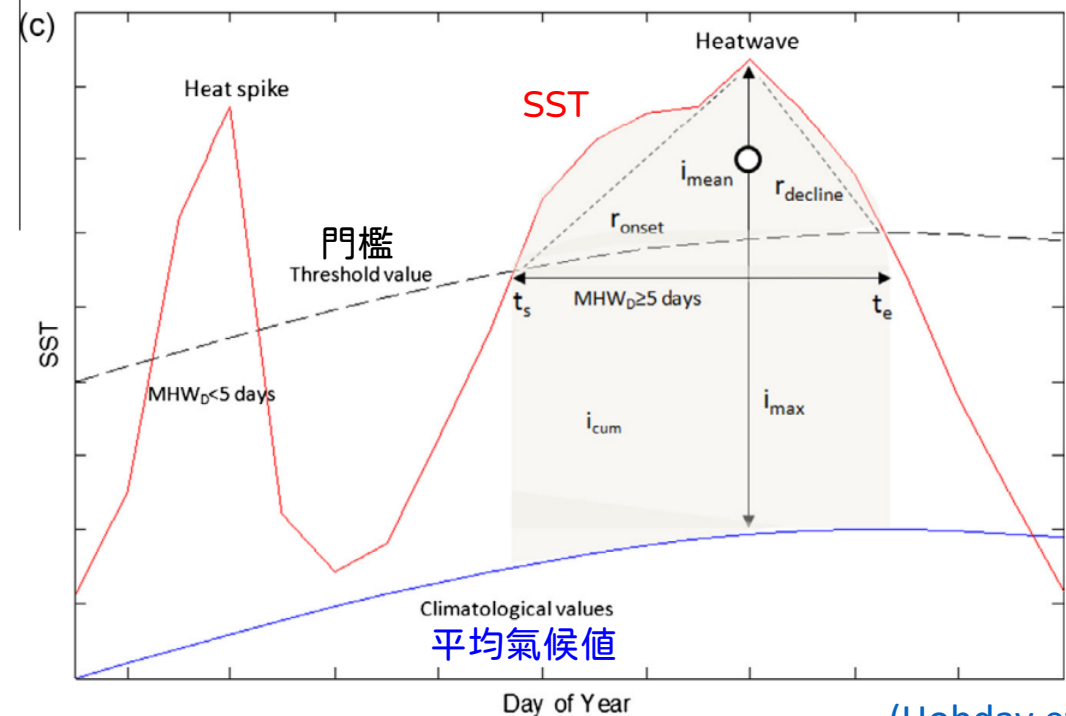
Received in revised form 31 December 2015

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Available online 6 January 2016

ABSTRACT

Marine heatwaves (MHWs) have been observed around the world and are expected to increase in intensity and frequency under anthropogenic climate change. A variety of impacts have been associated with these anomalous events, including shifts in species ranges, local extinctions and economic impacts on seafood industries through declines in important fishery species and impacts on aquaculture. Extreme temperatures are increasingly seen as important influences on biological systems, yet a consistent definition of MHWs does not exist. A clear definition will facilitate retrospective comparisons between MHWs, enabling the synthesis and a mechanistic understanding of the role of MHWs in marine ecosystems. Building on research into atmospheric heatwaves, we propose both a general and specific definition for MHWs, based on a hierarchy of metrics that allow for different data sets to be used in identifying MHWs. We generally define a MHW as a prolonged discrete anomalously warm water event that can be described by its duration, intensity, rate of evolution, and spatial extent. Specifically, we consider an anomalously warm event to be a MHW if it lasts for five or more days, with temperatures warmer than the 90th percentile based on a 30-year historical baseline period. This structure provides flexibility with



(Hobday et al., 2016)

• Marine Heatwave (Hobday et al., 2016)

- 日尺度
- 超過該地點、該日期的高溫門檻（90th percentile）
- 至少持續 5 天

(Hobday et al., 2016)

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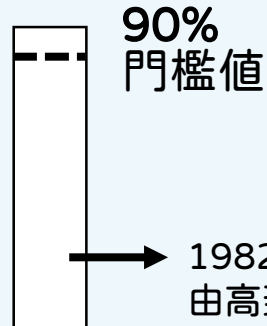
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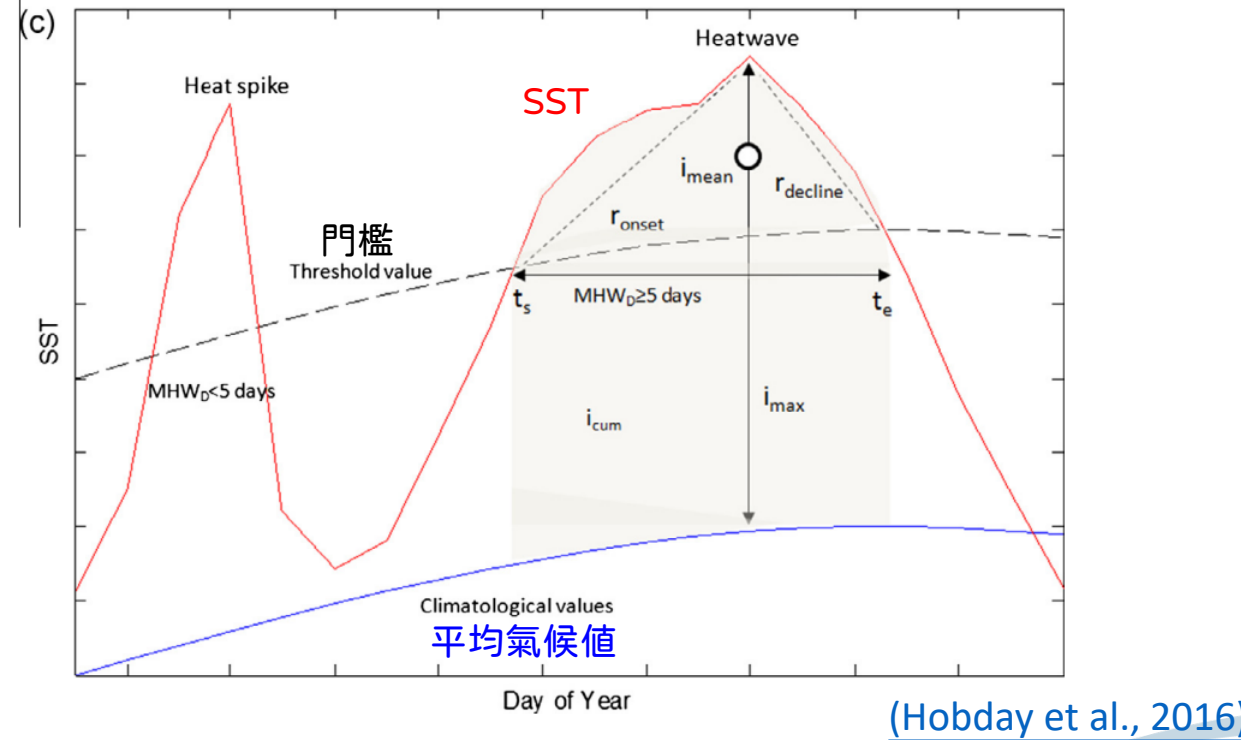
ABSTRACT

Marine heat
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90%
門檻值

1982-2011每年的距平值
由高到低排列



(Hobday et al., 2016)

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同一個概念，可以有不同時間尺度

Article

Thermal displacement by marine heatwaves

<https://doi.org/10.1038/s41586-020-2534-z>

Received: 28 January 2020

Accepted: 15 June 2020

Published online: 5 August 2020

 Check for updates

Michael G. Jacox^{1,2,3}✉, Michael A. Alexander², Steven J. Bograd^{1,2} & James D. Scott^{2,4}

Marine heatwaves (MHWs)—discrete but prolonged periods of anomalously warm ocean temperatures—can drastically alter ocean ecosystems, with profound ecological and socioeconomic impacts^{1–8}. Considerable effort has been directed at understanding the patterns, drivers and trends of MHWs globally^{9–11}. Typically, MHWs are characterized on the basis of their intensity and persistence at a given location—an approach that is particularly relevant for corals and other sessile organisms that must endure increased temperatures. However, many ecologically and commercially important marine species respond to environmental disruptions by relocating to favourable habitats, and dramatic range shifts of mobile marine species are among the conspicuous impacts of MHWs^{1,4,12,13}. Whereas spatial temperature shifts have been studied extensively in the context of long-term warming trends^{14–18}, they are unaccounted for in existing global MHW analyses. Here we introduce thermal displacement as a metric that characterizes MHWs by the spatial shifts of surface temperature contours, instead of by local temperature anomalies, and use an observation-based global sea surface temperature dataset to calculate thermal displacements for all MHWs from 1982 to 2019. We show that thermal displacements during MHWs vary from tens to thousands of kilometres across the world's oceans and do not correlate spatially with MHW intensity. Furthermore, short-term thermal displacements during MHWs are of comparable magnitude to century-scale shifts inferred from warming trends¹⁸, although their global spatial patterns are very different. These results expand our understanding of MHWs and their potential impacts on marine species, revealing which regions are most susceptible to thermal displacement, and how such shifts may change under projected ocean warming. The findings also highlight the need for marine resource management to account for MHW-driven spatial shifts, which are of comparable scale to those associated with long-term climate change and are already happening.

- 時間解析度→月
- 定義：每月海表溫度的距平值至少高於季節背景值的第 90 個百分位數。
- 距平值
- 季節背景值
- 第 90 個百分位數

同一個概念，可以有不同時間尺度

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季節背景值：將計算月份的前後一個月納入當作背景值

Ex. 想判斷 2 月是否有海洋熱浪

→ 綜合過去 1982-2011 年(氣候值)中 1、2、3 月全部每月 SST 距平值的第 90 個百分位數。



ODB MHW API 計算流程

月平均海溫



SST 距平值



門檻值



每月海洋熱浪

資料來源:

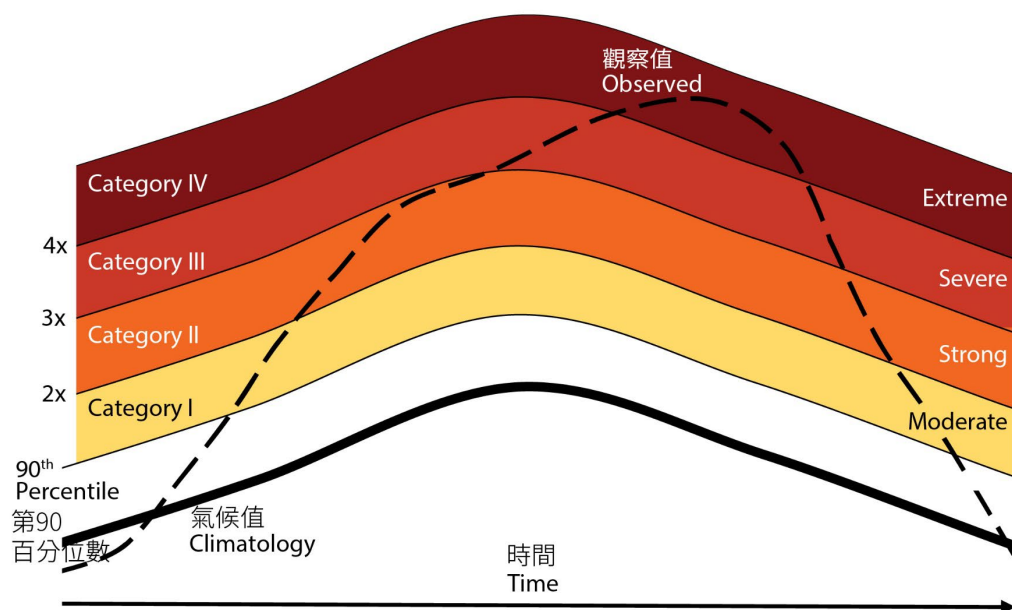
[NOAA OISST v2.1](#)

日尺度

0.25°

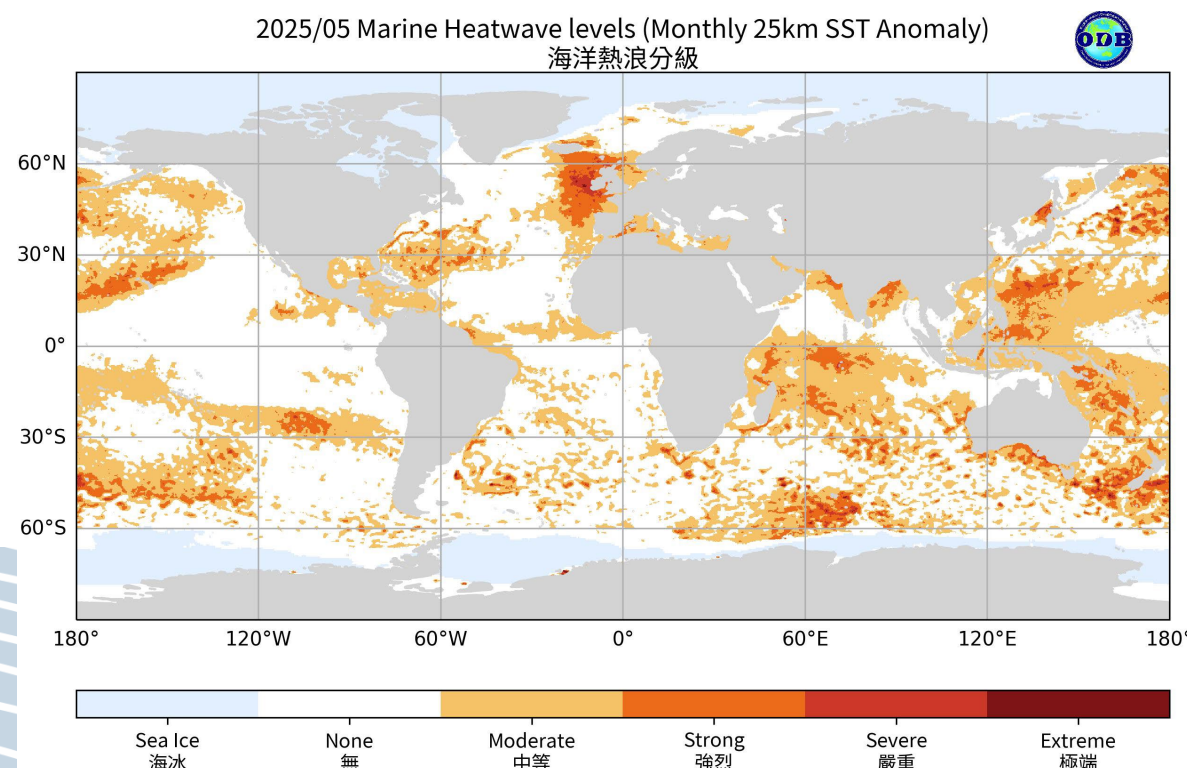
用 Daily SST 計算每
月平均溫度每月平均溫度
減
(1982~2011 氣
候背景值)高於季節背景值的
第 90 個百分位數
(Jacox et al., 2020)距平值 > 門檻值
→ 海洋熱浪Monthly product ≠
Hobday daily 5-day
event

描述海洋熱浪的嚴重度

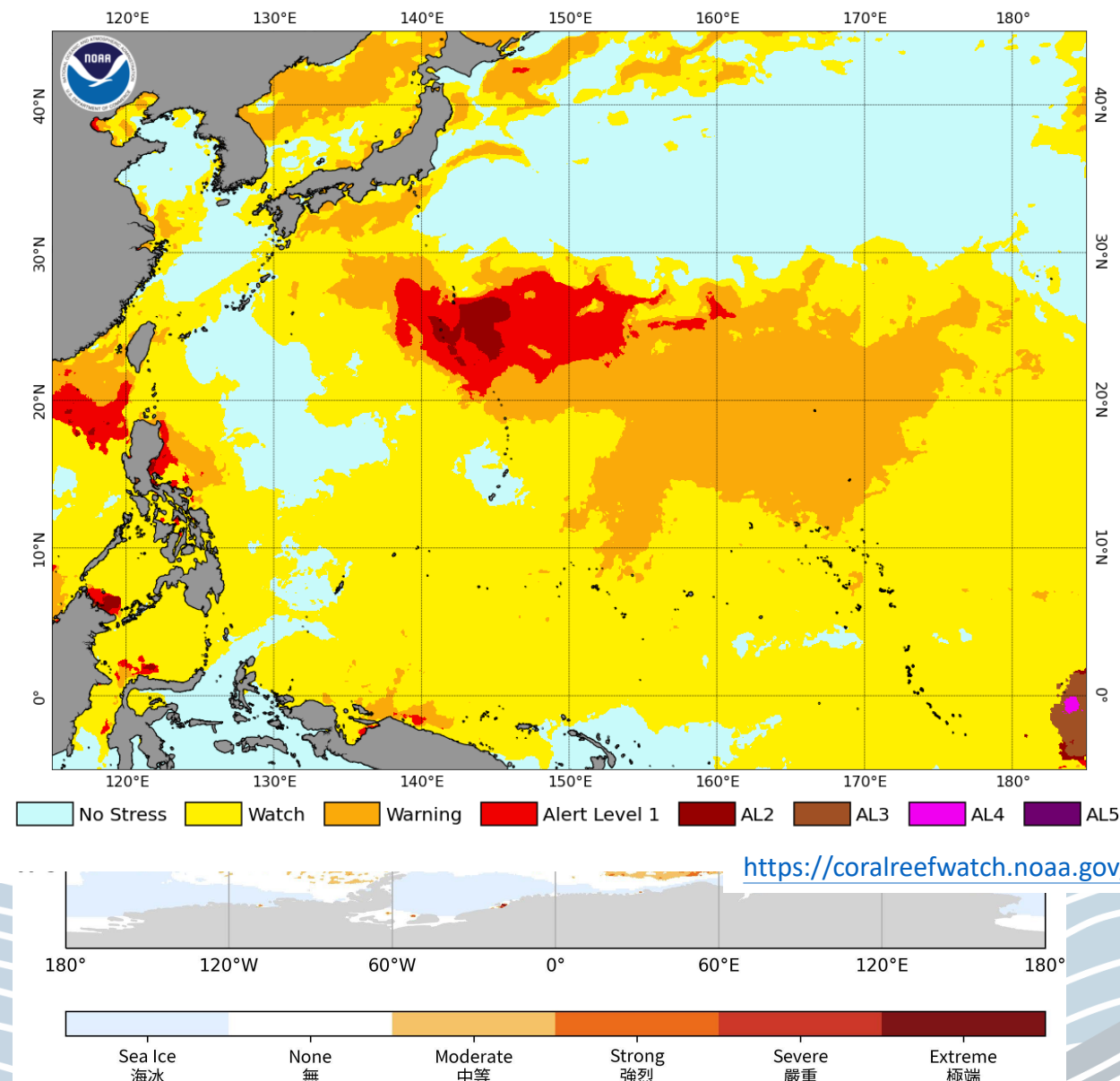
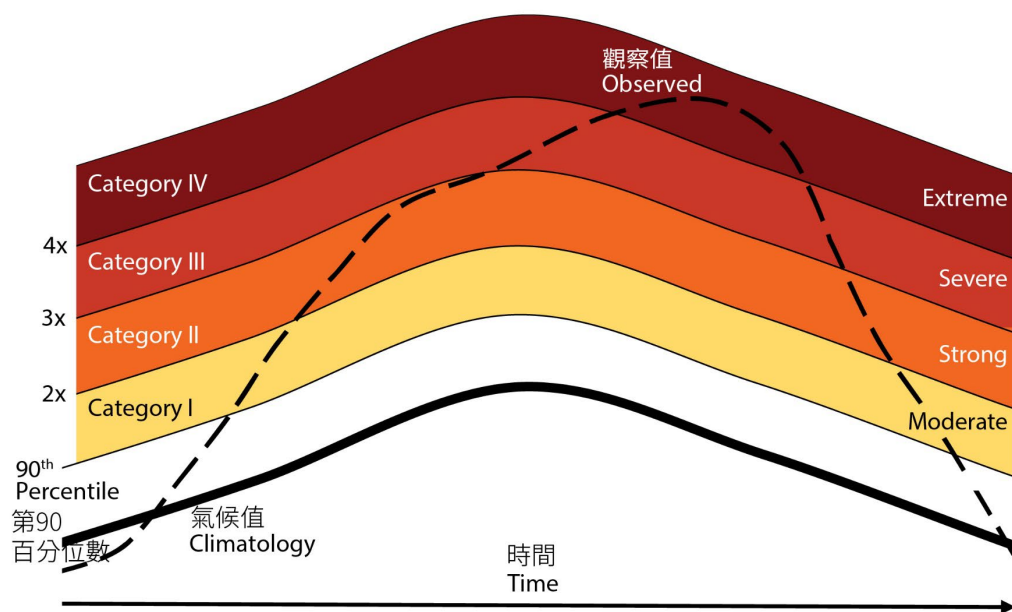


- 中等 (Moderate)
- 強烈 (Strong)
- 嚴重 (Severe)
- 極端 (Extreme)

2025/05 Marine Heatwave levels (Monthly 25km SST Anomaly)
海洋熱浪分級



描述海洋熱浪的嚴重度

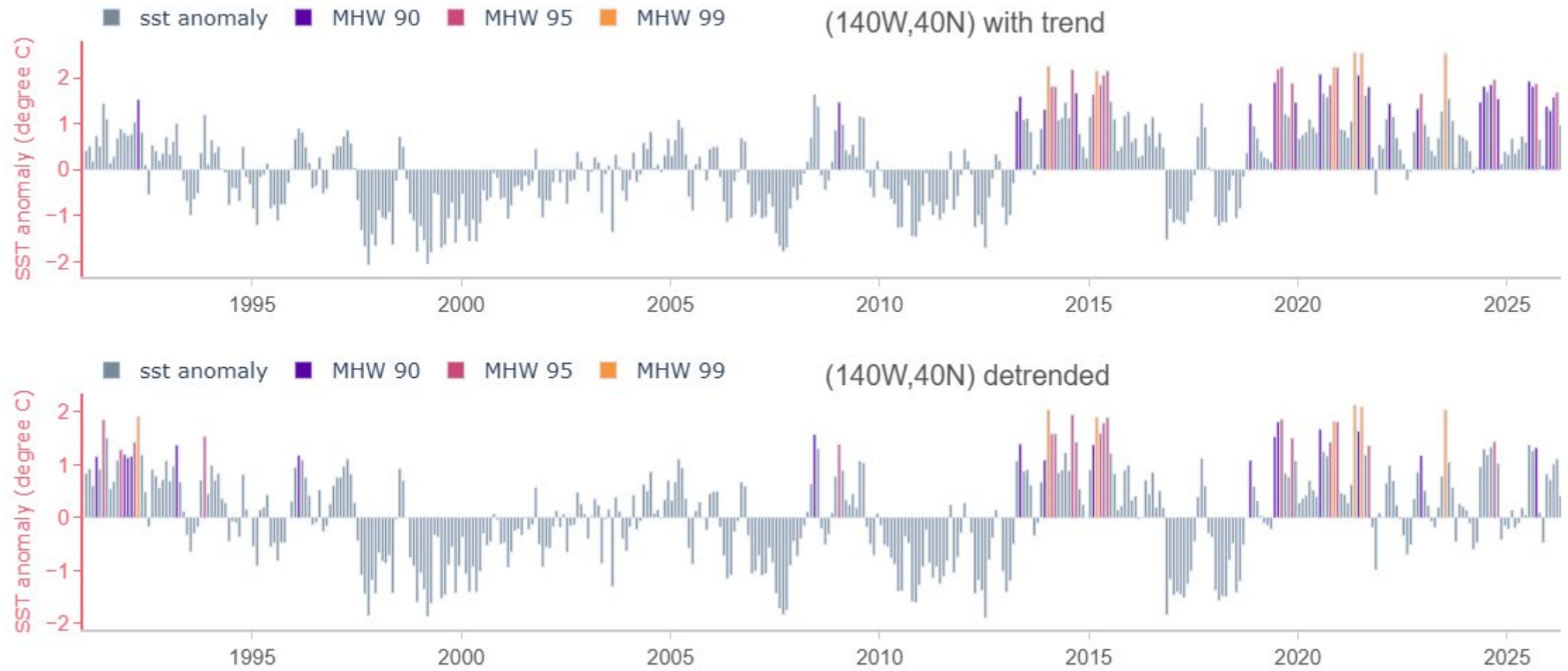


Hidy Viewer 

<https://odbview.oc.ntu.edu.tw/hidy/>

Coding Time

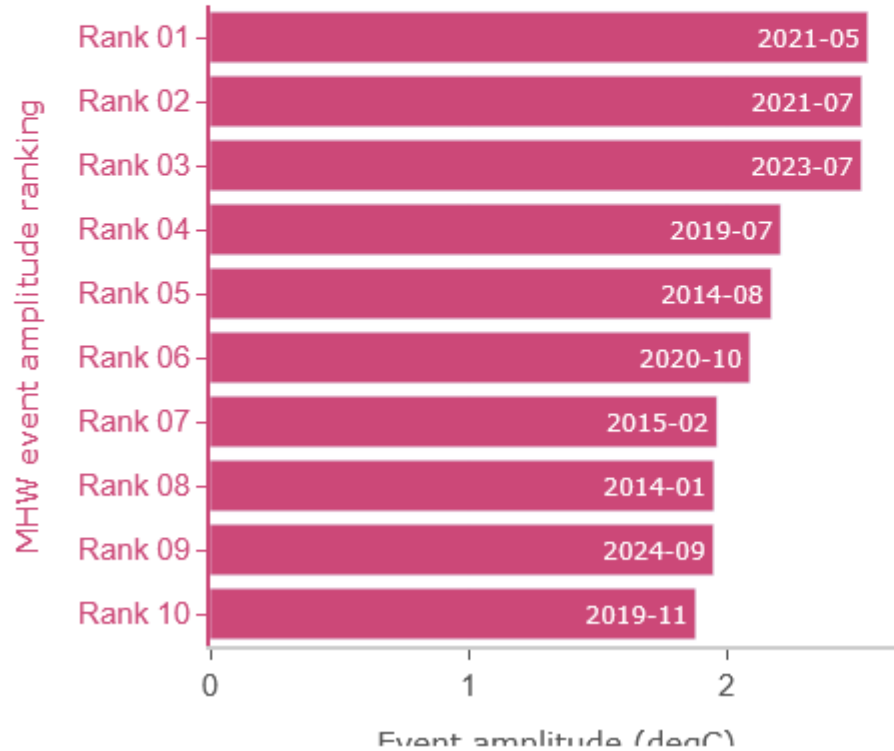




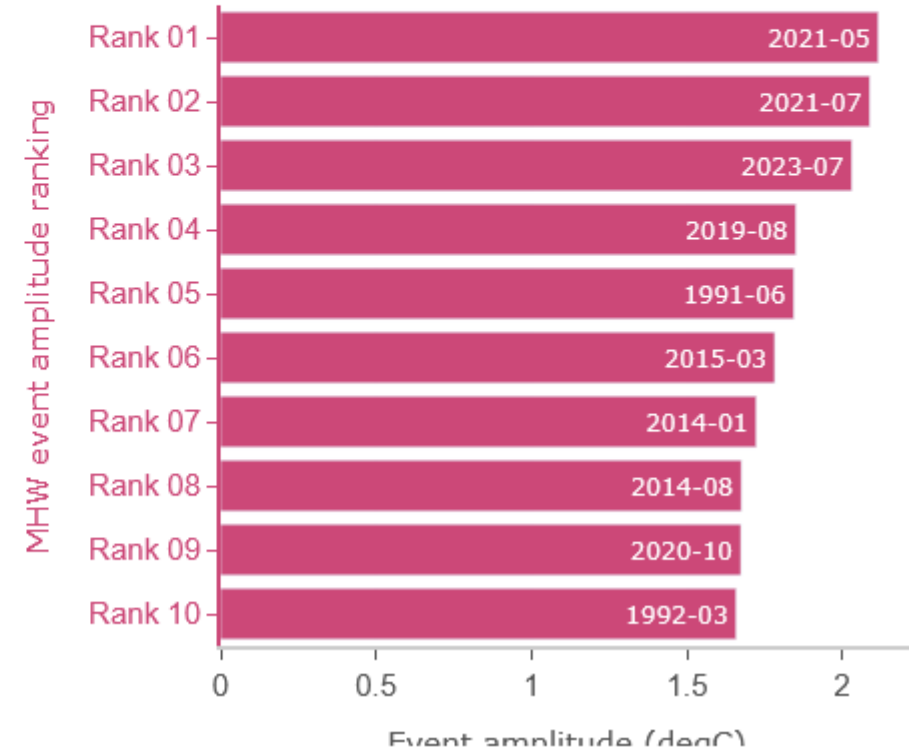
第95百分位數

Top 10 warmest MHW events

(140W,40N) with trend 長期暖化



(140W,40N) detrended 移除長期暖化



參考資料



- ODB 海洋熱浪
<https://eco.odb.ntu.edu.tw/pub/MHW/>
- Hidy Viewer
<https://odbview.oc.ntu.edu.tw/hidy/>
- NOAA 海溫資料 (OISST)
<https://www.ncei.noaa.gov/products/optimum-interpolation-sst>
- NOAA Physical Sciences Laboratory
<https://psl.noaa.gov/marine-heatwaves/>
- NOAA Coral Watch
<https://coralreefwatch.noaa.gov/>